

Project Initialization and Planning Phase

Date	26 June 2024
Team ID	SWTID-2026-1236
Project Title	Credit Card Approval Eligibility
Maximum Marks	5 Marks

Project Proposal report :

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective of this project is to develop an AI & ML-based web application that predicts whether a customer is eligible for a credit card. The system provides a quick and accurate eligibility prediction, helping users make informed decisions before applying for a credit card.
Scope	The project focuses on developing a web application where users can enter their personal and financial details. The machine learning model analyzes the input and predicts whether the customer is Accepted or Rejected for credit card eligibility.
Problem Statement	
Description	Many customers are not sure whether they meet the eligibility criteria for a credit card before applying. This uncertainty may lead to unnecessary application rejections and confusion because the approval criteria are not always clear.
Impact	Providing an AI-based eligibility prediction system helps customers understand their chances of approval before applying. It reduces uncertainty, saves time, improves decision-making, and enhances the overall user experience.
Proposed Solution	

Approach	Develop an AI & ML-based web application that analyzes customer details using a trained machine learning model and predicts whether the customer is likely to be Accepted or Rejected for credit card eligibility.
Key Features	• AI & ML-based credit card eligibility prediction. • User-friendly web interface for entering customer details. • Real-time prediction results within a few seconds. • Fast, simple, and reliable eligibility checking.

Resource Requirements :

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard CPU(Intel i5 or equivalent)
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	256 GB SSD or above
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Pandas,NumPy,Scikit-learn,Matplotlib,Seaborn,Joblib
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Data	Source, size, format	Kaggle Credit Card Approval Dataset(made some changes using LLM)